🛠️ AI Workshop — Prerequisites & Setup Guide

A one-day hands-on workshop with open-source, local LLMs

** ⚠️ This is a HANDS-ON coding workshop. Please complete ALL setup BEFORE the day. **

We run everything **locally and for free** (no paid API keys), which means you must download a few GB of tools + AI models in advance — **do NOT leave this for the colleges shared Wi-Fi.**

> ⏱️ ***Time needed:*** *~60–90 minutes (mostly downloads that run in the background)*

> 💰 ***Cost:*** *100% FREE — no paid API keys, no cloud bills*

💻 STEP 0 — Check Your Laptop Can Handle It (Check your laptop configuration before proceeding!)

Local LLMs are memory- and disk-hungry. Check your machine **before** anything else:

Resource	Minimum	Recommended	Notes
RAM	8 GB	16 GB+	On 8 GB, stick to ~3B models. 16 GB runs 7–8B comfortably.
Free disk	15 GB	30 GB+	Models are 1–8 GB each; Docker images add up.
CPU	4 cores (x86-64 / Apple Silicon)	8+ cores	Modern CPU runs small quantized models fine.
GPU	Optional	NVIDIA (CUDA) / Apple Silicon	Optional, but speeds up generation a lot.
OS	Win 10/11 64-bit · macOS 12+ · Ubuntu 20.04+	Latest stable	**64-bit only.**

> 👂 ***Low-spec laptop (8 GB RAM)?*** *You'll still complete the workshop — just use the smaller model (*`llama3.2:1b`*).* 👂 ***No suitable laptop at all?*** *Tell the co-ordinator sir (Surya Bam) in advance — pairing up may be possible.*

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1. 🐍 Python 3.12 or newer (Projects we do need Python.)

- **Download (official only):** https://www.python.org/downloads/
- ✅ **Windows — CRITICAL:** on the first install screen, **tick "Add python.exe to PATH"** before clicking *Install Now*.
- **macOS:** the python.org installer works; or use Homebrew: `brew install python@3.12`
- **Linux (Ubuntu):** `sudo apt update && sudo apt install python3.12 python3.12-venv python3-pip`

Verify — open a **new** PowerShell(windows) / terminal(macos/linux):

```
python --version        (Windows)
  python3 --version       (macOS / Linux)
  pip --version
```

You should see `Python 3.12.x` (or newer).

> ⚠️ ***Windows: typing `python` opens the Microsoft Store?*** *Turn off the alias: Settings → Apps → Advanced app settings → App execution aliases → disable the `python.exe` entries. Or reinstall with "Add to PATH" ticked.*

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2. 🐙 Git (Used to clone starter code and push your finished projects to your portfolio)

- **Windows:** https://git-scm.com/download/win (run installer, accept defaults)
- **macOS:** `git --version` (prompts to install) or `brew install git`
- **Linux:** `sudo apt install git`

Verify: Open terminal/Powershell `git --version`

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3. 🧩 Visual Studio Code (code editor we will use.)

- **Download:** https://code.visualstudio.com/
- **After installing, add extensions** (Extensions icon on the left → search → Install):
  - **Python** (by Microsoft) — required
  - **Pylance** (usually installs with Python) — recommended
  - **Docker** (by Microsoft) — optional

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4. 🦙 Ollama — run open-source LLMs locally ⭐ *(most important part of this workshop)*

Ollama lets you run powerful AI models **on your own laptop, offline, for free**. No API keys.

- **Download for your OS:** https://ollama.com/download
  - Windows: run the installer.
  - macOS: download the app, open it once.
  - Linux: `curl -fsSL https://ollama.com/install.sh | sh`

Verify Ollama is running in Powershell/terminal:

```
ollama --version
```

Ollama runs a background server at `http://localhost:11434`. You can confirm by opening that address in a browser — it should say *"Ollama is running"*.

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5. 📥 Pull the AI Models ⭐ *(DO THIS AT HOME — several GB!)*

The projects need **two** models. Pull them in advance on a fast connection:

```
**How to download:** Type below commands one by one in PowerShell(windows) / terminal(m
  acos/linux):

  ollama pull llama3.2
  ollama pull nomic-embed-text
```

Model	Purpose	Approx. size
`llama3.2`	Chat / generation (the "brain")	~2 GB
`nomic-embed-text`	Embeddings for RAG search	~275 MB

Test the chat model works:

```
ollama run llama3.2
  > Say hello in one short sentence.
  > /bye

  NOTE: `/bye` is to exit.
  You should get a reply in a few seconds.
```

> 💡 ***8 GB RAM / slow machine?*** *Also pull the tiny model as a fallback:* `ollama pull llama3.2:1b` *— and tell the instructor you'll use it.* ⚠️ *These downloads are the #1 thing people forget.* ***If you skip this, you'll spend the workshop staring at a download bar.***

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6. 🔗 GitHub Account

You'll push finished projects here — this becomes your **portfolio** that recruiters see.

- **Sign up (if you don't have account):** https://github.com/signup
- 💡 Choose a clean, professional username.

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7. 🐋 Docker Desktop *(Optional: You can skip this and still complete workshop projects. Only needed if you want to package your app to "run anywhere.")*

- **Download:** https://www.docker.com/products/docker-desktop/
- **Windows:** may require enabling **WSL 2** (the installer guides you).
- **Verify:** `docker --version` and `docker run hello-world`

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8. 🔑 GitHub Personal Access Token *(Optional)*

The Code Review Agent can review public repos without a token, but a token raises your GitHub API rate limit and allows private repos.

- **Create one:** GitHub → *Settings* → *Developer settings* → *Personal access tokens* → *Tokens (classic)* → *Generate new token*
  - Direct link: https://github.com/settings/tokens
  - Scope: `public_repo` (or `repo` for private repos) is enough.
- **Copy it now** — GitHub only shows it once. You'll paste it into a `.env` file (step 9).

> 🔐 ***Security:*** *treat this token like a password. Never commit it to Git or paste it into a chatbot. The projects load it from a local* `.env` *file that is git-ignored.*

✅ Quick Checklist (tick these off before the day)

- [ ] 0. Laptop meets the hardware bar above (8 GB RAM / 15 GB free disk minimum)

- [ ] 1. **Python 3.12+** installed and on PATH
- [ ] 2. **Git** installed
- [ ] 3. **Visual Studio Code** + Python extension
- [ ] 4. **Ollama** installed (runs local LLMs)
- [ ] 5. **Models pulled:** `llama3.2` and `nomic-embed-text` *(several GB — make it downloaded before session at home/college!)*
- [ ] 6. **GitHub account** created
- [ ] 7. *(Optional)* **Docker Desktop** installed
- [ ] 8. *(Optional)* **GitHub Personal Access Token** created (for the Code Review Agent)
- [ ] 9. Clone Workshop Repo https://github.com/silwalprabin/AI-Workshop-From-Academics-Projects-to-Career.git (OR download and unzip) to your laptop
- [ ] 10. Do **Readiness Test** (Step 10) & ensure everything passes ✅

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9. 📦 Get the Workshop Files & Install Dependencies

A. Clone Workshop Repo (OR download and unzip) the workshop shared by Er. Prabin Silwal.

```
git clone git@github.com:silwalprabin/AI-Workshop-From-Academics-Projects-to-Career.git

  OR

  git clone https://github.com/silwalprabin/AI-Workshop-From-Academics-Projects-to-Care
  er.git
```

Then goto project directory:

```
cd AI-Workshop-From-Academics-Projects-to-Career
```

B. Project 1 — AI Resume Assistant:

```
cd projects/resume_assistant

  # create & activate a virtual environment
  # Windows:
  python -m venv .venv
  .venv\Scripts\activate

  # macOS / Linux:
  python3 -m venv .venv
  source .venv/bin/activate

  pip install -r requirements.txt

  # Windows:
  copy .env.example .env
  # macOS / Linux
  cp .env.example .env

  # GO BACK BEFORE SETUP ANOTHER PROJECT
  cd ../..
```

C. Project 2 — GitHub Code Review Agent (new terminal or `deactivate` first):

```
cd projects/code_review_agent

  pip install -r requirements.txt

  # Windows:
  copy .env.example .env
  # macOS / Linux
  cp .env.example .env
```

(Optional) open `.env` and, if you made a token in step 8, set `GITHUB_TOKEN=your_token_here`.

> 💡 *The default `.env` values (`OLLAMA_HOST=http://localhost:11434`, `LLM_MODEL=llama3.2`, `EMBED_MODEL=nomic-embed-text`) already match the models you pulled — no changes needed for Project 1.*

🧪 STEP 10 — Readiness Test (run the right before workshop!)

Open a Powershell/terminal and confirm each line works:

1. `python --version` → shows **3.12.x** (or newer) ✅
2. `git --version` → shows a version ✅
3. `ollama --version` → shows a version ✅
4. `ollama run llama3.2 "hello"` → replies within a few seconds ✅
5. `ollama list` → shows **llama3.2** AND **nomic-embed-text** ✅
6. In `projects/resume_assistant` with the venv active: `python -c "import streamlit, langgraph, chromadb; print('deps ok')"` → prints `deps ok` ✅
7. *(If doing Docker)* `docker run hello-world` → prints a success message ✅

If all pass, you're 100% ready to build. 🎉

🆘 Troubleshooting (quick fixes)

Problem	Fix
`python` not recognized (Windows)	Reinstall Python with **"Add to PATH"** ticked, then open a **new** terminal.
`python` opens Microsoft Store	Disable the Store alias in *App execution aliases*.
`ollama: command not found`	Reinstall Ollama; reopen the terminal; ensure the Ollama app is running.
Model download is very slow	Do it at home on fast Wi-Fi; it resumes if interrupted. Try `llama3.2:1b`.
`pip install` fails on a package	Ensure your **venv is active** and Python is 3.12+; upgrade pip: `python -m pip install --upgrade pip`.
"Connection refused" to Ollama	Make sure Ollama is running (open http://localhost:11434 in a browser).
Out of memory / very slow generation	Use a smaller model: `ollama pull llama3.2:1b` and set `LLM_MODEL=llama3.2:1b` in `.env`.
Docker won't start (Windows)	Enable **WSL 2** and virtualization in BIOS; restart.

> 📄 *A **full** troubleshooting guide is in the workshop repo folder (AI-Workshop-From-Academics-Projects-to-Career): `docs/TROUBLESHOOTING.md`. The complete per-OS install reference is in `docs/INSTALLATION_GUIDE.md`.*

📋 What to Bring on Workshop Day

- 💻 Your laptop, **fully charged**, + charger 🔌
- ✅ **All setup above completed and the Readiness Test passing** (this is the big one!)
- 🦙 Both models already pulled (`llama3.2` + `nomic-embed-text`)
- 🔑 GitHub login ready (+ token if you made one)
- 📓 A notebook or note-taking app
- 🧠 Curiosity and questions — **no question is too basic!**

⏰ Why doing this in advance matters

This is a **build day**, not a lecture. Every minute spent downloading Python, Ollama, or a 2 GB model on shared venue Wi-Fi is a minute you're *not* building your portfolio projects. Students who arrive set up finish **both** projects and leave with GitHub-ready work. **Please come ready.** 🚀

See you at the workshop! You're about to build two real AI projects that run entirely on your own machine — for free.

> ℹ️ *Tools and links change over time. If a link breaks, search the tool's official name (Python, Git, VS Code, Ollama, Docker) — the official site is the first result.*

Thank you.

Your Instructor

Er. Prabin Silwal

- LinkedIn: https://www.linkedin.com/in/prabin-silwal/
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